TechPad Case: FreeCAD Build Guide

Three-piece 3D-printed key-well case for the TechPad PCB (92 × 87 mm). Every number here comes from your routed board file (pcb.kicad_pcb, 18 Oct 2026). Units are mm. Written for FreeCAD 1.0. Version 2, 19 Oct 2026: key-well design.

Confidence tags: Certain checked against your PCB file or the official rules. Likely standard practice, not tested on your parts yet. Guessing a typical value you must confirm. MEASURE a typical value you can't check until the parts arrive. Model it as written now and fix it later (see 05_After_Parts_Arrive).

Read these four things before you start

1. **You design now and build later.** Certain Hack Club sends the parts after you submit, so you can't measure anything or test-print yet. Model the case with the numbers in this guide exactly as written. Everything that depends on the real parts is in **05_After_Parts_Arrive**.
2. **FreeCAD is a good fit for this.** Certain It opens DXF files and your KiCad STEP file directly, and it saves STEP itself, so you don't need any conversion tricks.
3. **This guide uses the Part workbench, not Part Design.** Likely The DXF drawings already contain every outline at the right position *and the right height*. So every feature is the same 2 clicks: **Extrude** an outline, then **Cut** it out or **Union** it on. You don't draw any sketches or constraints.
4. **Each of the 3 pieces prints without supports.** Likely That's why the case is split this way: a flat plate, a frame printed face-down, and a tray printed floor-down.

1. What you are building

- **Top frame.** This is the part you see. It's a 6.4 mm thick frame with one opening (the *key well*) around all 12 keys. It reaches to 1 mm under the keycaps, so the switch bodies stay hidden. It also has the OLED window, the knob hole, the logo, the border groove and the 4 hidden heat-set inserts underneath.
- **Switch plate.** A flat 1.5 mm plate. The 12 switches clip into it. Four small bosses underneath stop 0.2 mm above the PCB, around the screws.
- **Bottom tray.** The PCB sits inside it on 4 posts. The 4 screws come in from underneath, and there's a USB-C slot in the back wall.
- **Knob cap.** A small printed cap for the EC11 encoder shaft.

Outside size: **98 × 93 × 22.4 mm**, with 8 mm rounded corners. That's the case only; the keycaps stick out about 8 mm above the frame. Hackpad's limit is 200 × 200 × 100. Certain

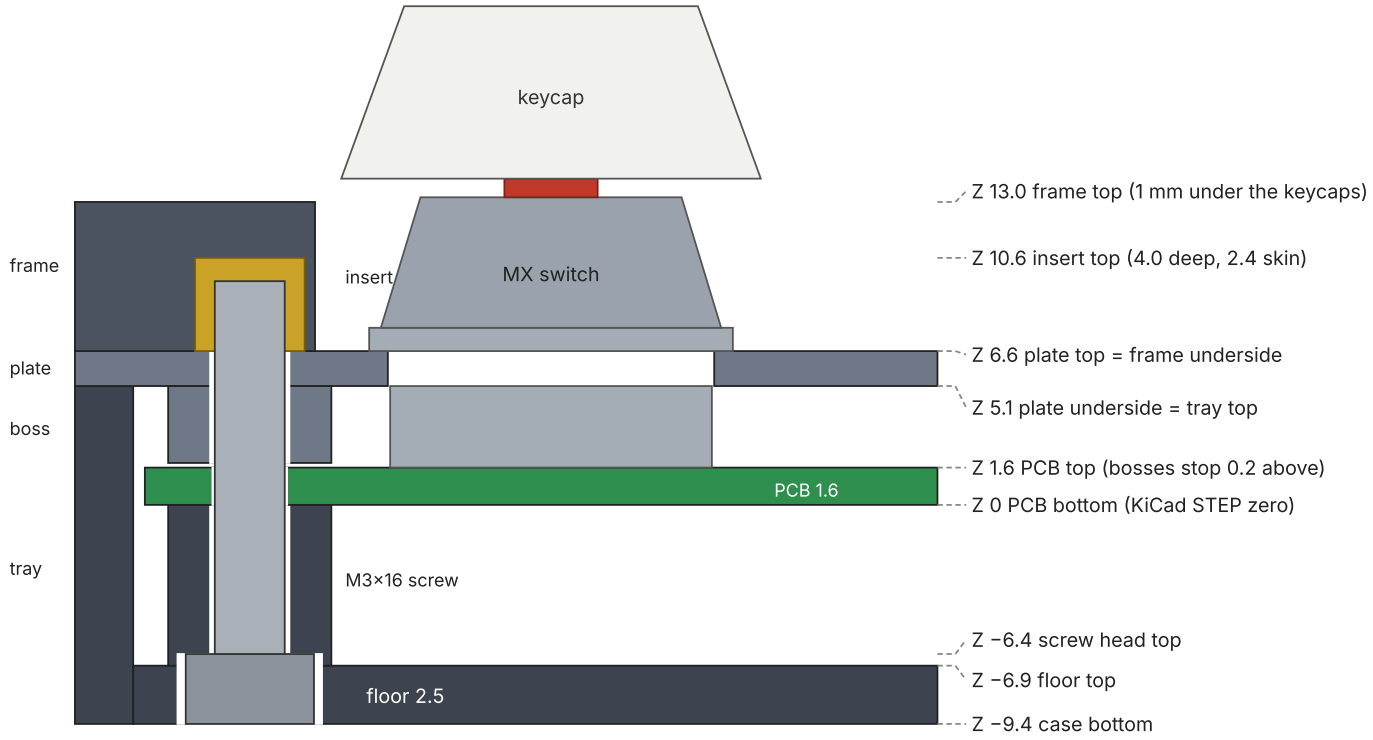


The finished look (3D preview: https://claude.ai/artifact/4LtL77XX4mXD5pFEnwg2R4).

2. Coordinates: the most important page

Work in one coordinate system the whole time. The PCB, the DXF drawings and the case will then line up by themselves.

- **X/Y origin (0,0) = bottom-left corner of the PCB**, seen from the top. X goes right, Y goes up. The PCB runs from (0,0) to (92,87).
- **Z = 0 at the PCB's bottom face.** That's the same zero KiCad uses when it exports a STEP file. The PCB top is at Z = 1.6.
- To turn a KiCad position into a case position: X = KiCad X − 100, Y = 187 − KiCad Y. For example, KiCad (117.425, 133.525) is case (17.425, 53.475).



Cross-section through a corner screw and the bottom-left switch. Each M3×16 screw goes up through the tray, the PCB and the plate into a heat-set insert in the frame, so no screws show on top.

Height stack (Z levels)

Z (mm)	What is there	Why
13.0	Frame top (the top surface of the case)	Keycaps rest at Z 14.0, so the frame stops 1 mm under them and hides the switch bodies. Likely
6.6 to 10.6	Heat-set insert hole, Ø4.7 × 4.0 deep, under each frame corner	Uses the M3×5×4 inserts from your kit; leaves a 2.4 mm skin above. Likely
6.6	Plate top = frame underside	MX standard: the plate top is 5.0 mm above the PCB top. Certain
5.1 to 6.6	Switch plate, 1.5 mm thick	Switch clips need exactly 1.5 mm. Certain
5.1	Tray wall top = plate underside	The plate sits on the tray walls.
1.8 to 5.1	Plate bosses, Ø7, at the 4 screw holes	They stop 0.2 mm above the PCB, so print errors can't rock the plate. The soldered switches already hold the PCB. Likely
0	PCB bottom, resting on the 4 posts	Same zero as the KiCad STEP export.
about −1.0 to −4.2	XIAO hanging under the PCB, USB-C centre at about −2.6	Guessing MEASURE
−5.6 to 0.4	USB-C slot in the back wall, 12 × 6	Room for a normal USB-C cable plug. MEASURE
−6.4	Screw head top (Ø6.2 × 3.0 counterbore)	Puts the tip of a 16 mm screw at Z 9.6, which is 3.0 mm into the insert. Certain (arithmetic)
−6.9	Floor top	Leaves 2.7 mm under the XIAO.
−9.4	Case bottom	The 2.5 mm floor.

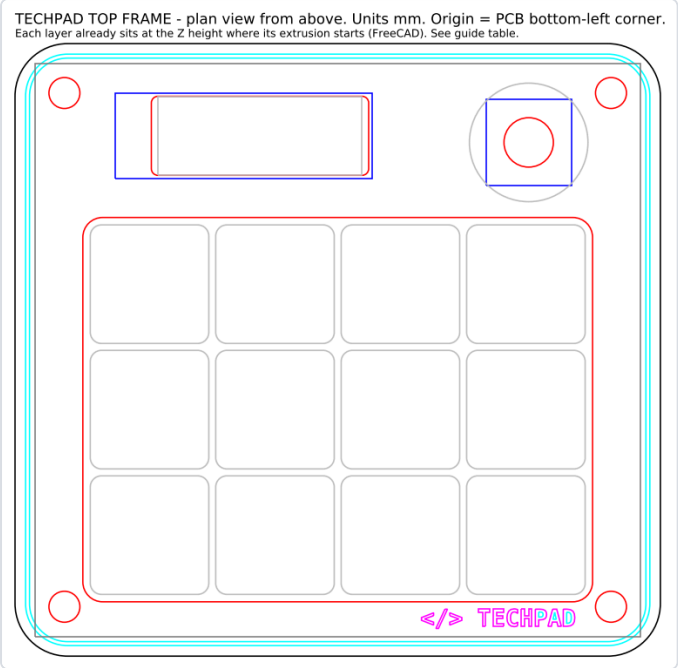
3. Typical values used for the parts MEASURE

The PCB numbers are exact. These part numbers are typical values, and real parts vary between batches. **Don't wait on them: model with these values now.** When the parts arrive, measure them and make the fixes in 05_After_Parts_Arrive. Each fix is one DXF shape or one number, so it's a small change.

Part	Value used
OLED glass top above the PCB top	≈ 5.0, so it sits at the plate top (Z 6.6), at the bottom of the frame window
OLED glass outline	31 × 12, X 18.7 to 49.7, Y 70 to 82. The frame window is 1 mm bigger at each end
OLED module board	38 × 12, X 12.7 to 50.7. The plate has a 39 × 13 cut for it
Encoder body above the PCB	≈ 6.5 (top at Z 8.1). The frame has a 2 mm pocket underneath for it
Encoder bushing and shaft	Ø7 bushing; Ø6 D-shaft, 4.5 across the flat, top at Z ≈ 21.6
XIAO USB-C centre	Z ≈ −2.6
Heat-set insert	5.0 × 4.0, hole Ø4.7 × 4.0
Screw	M3×16, socket head Ø5.5 × 3.0

4. Files you will use

File (in drawings/)	What it is
TechPad_Frame_plan.dxf	Every top-frame outline, one per layer, at the true X/Y position and already at the Z height where its extrusion starts.
TechPad_Plate_plan.dxf	Plate outline, switch holes, encoder and OLED cuts, bosses and screw holes.
TechPad_Tray_plan.dxf	Tray walls, posts, screw holes, counterbores, USB slot.
TechPad_PCB_reference.dxf	PCB outline, holes, switch centres, encoder, OLED header pins, XIAO footprint.
pcb.step (you export it, step 5.4)	Your real board in 3D, for checking fit.



Frame DXF

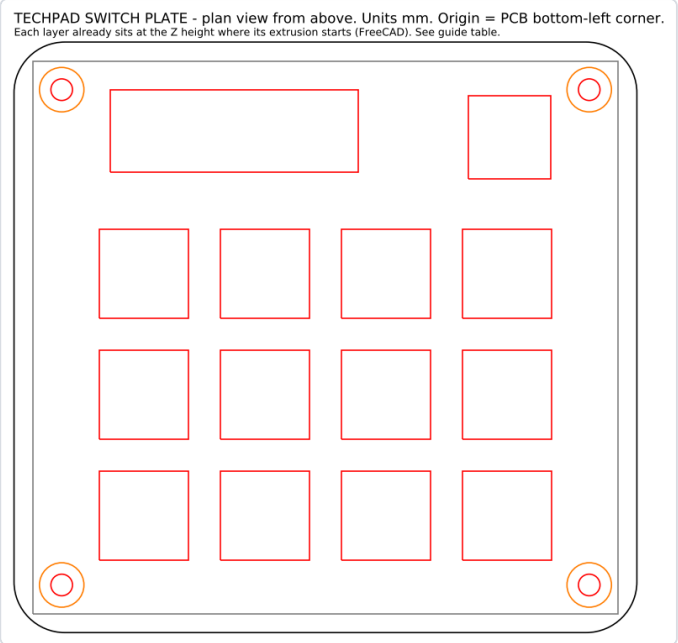
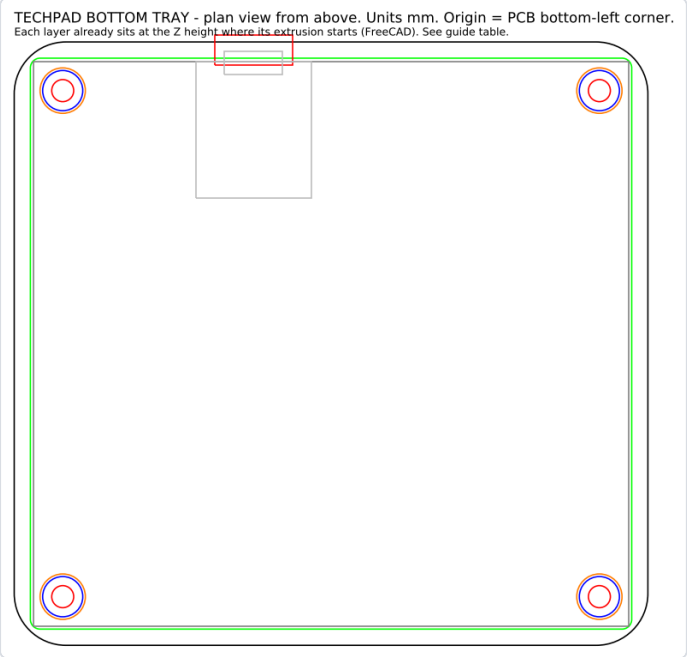


Plate DXF



Tray DXF

5. FreeCAD setup (once)

5.1 Open FreeCAD, start a **New** document and save it as **TechPad_Case.FCStd**. Switch to the **Part** workbench using the workbench drop-down in the toolbar.

5.2 Import the drawings: **File → Import** → TechPad_Frame_plan.dxf, then the same for TechPad_Plate_plan.dxf and TechPad_Tray_plan.dxf. Likely

- Each DXF layer shows up in the tree on the left as its own layer group, with the layer names from the tables below.
- Use the **Space bar** on a selected group to hide or show it. Keep only the layers you need visible.
- Tip: build one piece at a time and hide the other two DXFs while you do.

5.3 **Check the heights.** Press **0** (isometric view). The outlines should float at different heights: the logo and groove at the top, the tray outline far below.

- If everything is flat at one height, your FreeCAD ignored the DXF heights. Then, for each layer group, select its objects and set **Placement → Position → z** in the Property view (Data tab) to the "Start Z" value in the tables.

5.4 Bring in your real board to check fit:

- In KiCad's PCB editor go to **File → Export → STEP**. Set the origin to **User defined**, X = 100, Y = 187, and save it as pcb.step. That puts KiCad's (100,187) at the PCB's bottom-left corner, which is our 0,0.
- In FreeCAD, **File → Import** → pcb.step. It should land exactly inside the case outlines.
- If it doesn't load, make a stand-in: Part → Primitives → **Cube**, 92 × 87 × 1.6 at position 0,0,0.

5.5 The 2-click recipe used for every feature in sections 6 to 8:

1. In the tree, select the outlines on the named layer. Ctrl+click picks several; Shift+click picks a range.
2. **Part → Extrude**. Set Direction to **Custom direction** with X 0, Y 0, Z 1. Set **Length → Along** to the table value, and **tick "Create solid"**. Click OK.
3. Combine:
   - **Part → Boolean → Cut**: select the part first, then Ctrl+click the tool, then Cut. Cut takes exactly 2 objects, so when there are several tool solids (for example the 12 switch squares), first select them all and use **Part → Boolean → Union** to fuse them into one tool.
   - **Part → Boolean → Union**: adds solids to the part.

Every cutting tool is 0.1 mm longer than it needs to be. That avoids paper-thin leftover faces. Keep the tree tidy by renaming each result with F2 (for example "Frame_after_F4").

6. Top frame, step by step (TechPad_Frame_plan.dxf)

#	DXF layer	Shape	Start Z (in DXF)	Extrude length	Then	Result
F1	FRAME_OUTLINE	98 × 93, R8 corners	6.6	6.4	-	Solid block, Z 6.6 to 13.0
F2	CUT_WELL	77.4 × 58.35, R3, X 7.3 to 84.7, Y 5.25 to 63.6	6.5	6.6	Cut	The key well, 0.6 mm round the keycaps
F3	CUT_OLED_WINDOW	33 × 12, R1, X 17.7 to 50.7, Y 70 to 82	6.5	6.6	Cut	OLED window **MEASURE**
F4	POCKET_ENCODER_UNDERSIDE	13 × 13 square at (75, 75)	6.5	2.1	Cut	Pocket for the encoder body, Z 6.6 to 8.6 **MEASURE**
F4b	POCKET_OLED_UNDERSIDE	39 × 13, X 12.2 to 51.2, Y 69.5 to 82.5	6.5	2.5	Cut	Pocket over the OLED module, Z 6.6 to 9.0, so you can solder the screen up to 3.5 mm higher **Likely**
F5	CUT_ENCODER_HOLE	Ø7.5 at (75, 75)	6.5	6.6	Cut	Hole for the bushing and shaft
F6	INSERT_HOLE	4 × Ø4.7	6.5	4.1	Union the 4, then Cut	Insert holes 4.0 deep, 2.4 skin above
F7	GROOVE_TOP	2 loops: inset 1.6 and 2.2	12.6	0.5	see below	0.6 wide × 0.4 deep border groove
F8	LOGO + LOGO_HOLES	"</> TECHPAD", 2.6 tall	12.6	0.5	see below	0.4 deep engraved logo

> **F7 groove:** Extrude both loops. Then Cut the *inner* solid out of the *outer* one, which leaves a thin ring. Cut the ring from the frame.

> **F8 logo:**
>
> 1. Extrude all LOGO outlines (10 letters and symbols) and all LOGO_HOLES outlines (the insides of P, A and D).
> 2. Union the 10 letters into one solid and the 3 holes into another, then Cut the holes out of the letters.
> 3. Cut the finished logo from the frame.

> **F9 edges:** **Part → Chamfer**. Select the frame, tick the 8 edges around the outer **top** edge, length 0.6. Then chamfer the 8 outer **bottom** edges by 0.5. Optional: chamfer the top edge of the key well by 0.5.

> **F10 screen bezel:** **Part → Chamfer** on the 8 top edges of the OLED window (4 straight, 4 corner arcs), length **3.0**. That turns the top 3 mm of the window into a 45° slope, so the opening is 39 × 18 at the top and narrows to the 33 × 12 window 3.4 mm above the glass. It makes the screen look framed rather than sunk in a slot, and lets you read it from a lower angle. If FreeCAD refuses 3.0, try 2.5. Likely

Check: Turn on REF_KEYCAP, REF_KNOB and REF_OLED_GLASS. Every keycap outline should sit inside the well with a gap all round, and the glass outline should sit inside the window. The tightest spot is the well's corners, where the gap is about 0.3 mm. That's fine, because the caps move straight up and down. Certain

7. Switch plate, step by step (TechPad_Plate_plan.dxf)

#	DXF layer	Shape	Start Z (in DXF)	Extrude length	Then	Result
P1	PLATE_OUTLINE	98 × 93, R8	5.1	1.5	-	Flat plate, Z 5.1 to 6.6
P2	CUT_SWITCH	12 × 14.0 squares	5.0	1.7	Union the 12, then Cut	Switch holes
P3	CUT_ENCODER	13 × 13 at (75, 75)	5.0	1.7	Cut	Encoder body passes through
P4	CUT_OLED_MODULE	39 × 13, X 12.2 to 51.2, Y 69.5 to 82.5	5.0	1.7	Cut	The whole OLED module passes through MEASURE
P5	BOSS	4 × Ø7	1.8	3.4	Union	Bosses under the plate, stopping 0.2 above the PCB
P6	SCREW_HOLE	4 × Ø3.4	1.5	5.2	Union the 4, then Cut	Screw clearance through bosses and plate

> **P7 edges:** **Part → Chamfer** 0.5 on the 8 outer top edges and the 8 outer bottom edges. With the chamfers on the frame and the tray, that gives two fine V-lines on the side, with the plate showing as a thin stripe between them.

8. Bottom tray, step by step (TechPad_Tray_plan.dxf)

#	DXF layer	Shape	Start Z (in DXF)	Extrude length	Then	Result
B1	TRAY_OUTLINE	98 × 93, R8	−9.4	14.5	-	Block, Z −9.4 to 5.1
B2	TRAY_CAVITY	93 × 88, R1.5	−6.9	12.1	Cut	Tray with a 2.5 floor and 2.5 walls
B3	POST	4 × Ø7	−7.0	7.0	Union	PCB support posts, tops at Z 0
B4	SCREW_HOLE	4 × Ø3.4	−9.5	10	Union the 4, then Cut	Screw clearance holes

B5	COUNTERBORE_UNDERSIDE	4 × Ø6.2	−9.5	3.1	Union the 4, then Cut	Head pockets, flush with the bottom
B6	USB_SLOT	12 × 4.5 rectangle, X 28 to 40	−5.6	6.0	Cut	Closed USB-C slot, Z −5.6 to 0.4 MEASURE

> B7 **Part → Fillet**, R1.5, on the 4 edges running front-to-back inside the USB slot, which gives it round corners like the plug. Then use **Part → Chamfer**: 0.5 on the outer top edge and 0.6 on the outer bottom edge.

> B8 Optional: four Ø10 × 0.8 recesses in the bottom near the corners for stick-on rubber feet. They aren't in the kit, so skip them unless you buy feet.

9. Knob cap

1. **Part → Primitives → Cylinder**: radius 9, height 12, position 75, 75, 14.5. The knob sits 1.5 mm above the frame. Likely
2. Bore: another Cylinder, radius 3.1, height 7.4, same position. Cut it from the knob. The shaft top is at Z ≈ 21.6, so the bore reaches 21.9.
3. D-flat: **Primitives → Box**, length 6.2, width 1.6, height 7.4, at position 71.9, 76.6, 14.5. Union it back into the knob. This leaves 4.6 across the flat, which is the shaft's 4.5 plus 0.1 so it pushes on. MEASURE
4. **Part → Fillet** 1.0 on the top edge. Optional: a 0.6 × 0.4 indicator line on top.

10. Check the fit in FreeCAD

- **Part → Check geometry** on each finished part. It should report one valid solid with no errors.
- **Interference:** make a *copy* of a part (Edit → Duplicate selection), select it and pcb.step, then **Part → Boolean → Intersection** (Common). The result should be empty for the tray and frame. For the plate there should be no overlap; the bosses stop 0.2 mm above the PCB.
- **Pieces stack:** the tray top, plate bottom, plate top and frame bottom should meet exactly at Z 5.1 and 6.6. Check them with **Part → Cross-sections**, plane XZ, at Y = 4.5. It should look like the cross-section in section 2.
- **No test print yet.** The fit tests are in 05_After_Parts_Arrive, because you need a real switch and insert to check them.

11. Export for printing and for Hackpad

File	How	Goes in
Frame.step, Plate.step, Tray.step, Knob.step	Select one finished part → **File → Export** → type **STEP with colors (*.step *.stp)**. Repeat for each part. Certain	Production/
Frame.stl, Plate.stl, Tray.stl, Knob.stl	Select one part → File → Export → **STL Mesh (*.stl)**. This is what your slicer prints. Likely	Production/ (optional extra)
TechPad.step (full assembly)	Select all 4 parts **and** pcb.step together → File → Export → STEP. Certain	CAD/ (must include the PCB; a blank rectangle is fine)

TechPad_Case.FCStd	Your source file.	CAD/

12. Printing and assembly: after the parts arrive

Print settings, the fit tests, the order to solder and screw it together, and the fixes for any part that measures differently are all in **05_After_Parts_Arrive**. For submitting, you only need the STEP files from section 11.

13. Common problems

Problem	Cause and fix
Extrude makes a face or a shell, not a solid	"Create solid" was not ticked, or the outline is not closed. Delete it and extrude again with the box ticked.
Cut is greyed out	Cut needs exactly 2 objects: the part first, then the tool. Union several tools into one first.
Outlines all import at Z 0	Your DXF importer ignored the heights. Set Placement → z on each layer, as in step 5.3.
Chamfer fails	Pick fewer edges at once, or chamfer before cutting the groove and logo.
Switches are loose or won't clip in	The plate isn't 1.5 thick, or the hole isn't 14.0. Calibrate your printer first, then change the hole by ±0.05.
Insert goes in crooked	The hole is too small, or the iron is too cold. Try Ø4.8 and go slowly.
Pieces gap at the corners	Warping. Use a brim, and print slower on the first layers.

Sources: Hackpad submission rules at blueprint.hackclub.com/hackpad/submitting. PCB numbers come from your pcb.kicad_pcb. Menu names are from FreeCAD 1.0. Nothing in this guide has been printed or test-fitted yet.